DOCUMENT CONTROL & METADATA			
Project Name:	Wing Structural Analysis Prototype Demo	**Lead Analyst:**	AVX Engineering Team
Simulation Type:	FEA	**Software Tool:**	SolidWorks 2026
Revision:	v1.0	**Date**	Oct. 05, 2026

1. Simulation Setup

SolidWorks FEA	
Analysis Type	Static
Unit System	SI (m-kg-s)
Total Cell Counts Mesh	532,674
Mesh Size	5.55 mm
Gravity	On
Calculation Time	2 Hours
Boundary Conditions	
Applied Force	5 N
Global Interaction	Bonded
Materials	PETG, Carbon Fiber, & Plywood

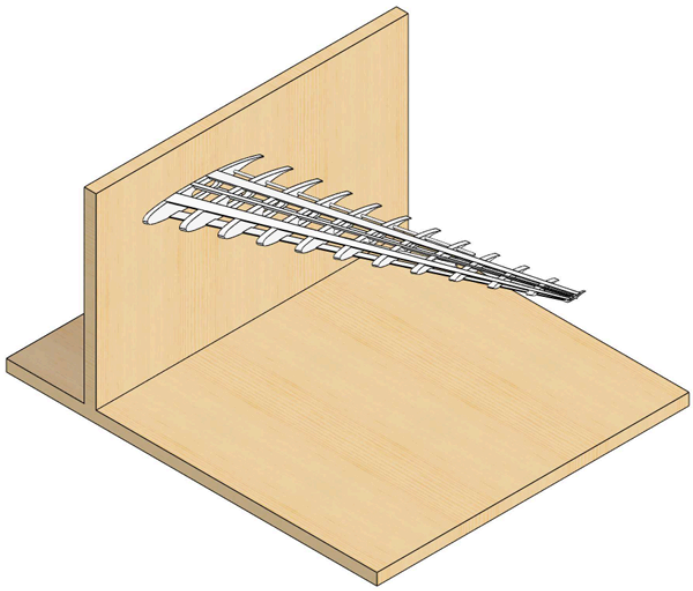


Figure 1. Isometric View

2. Simulation Report

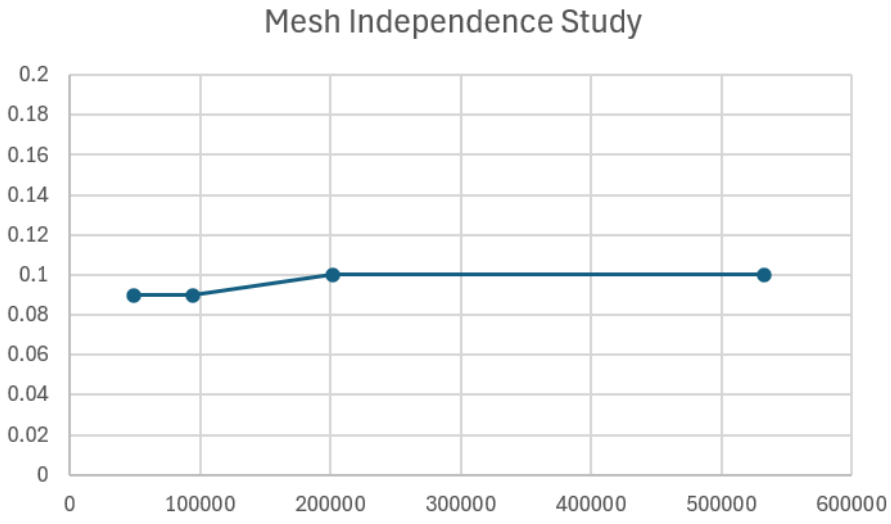


Figure 2. Mesh Independence Study

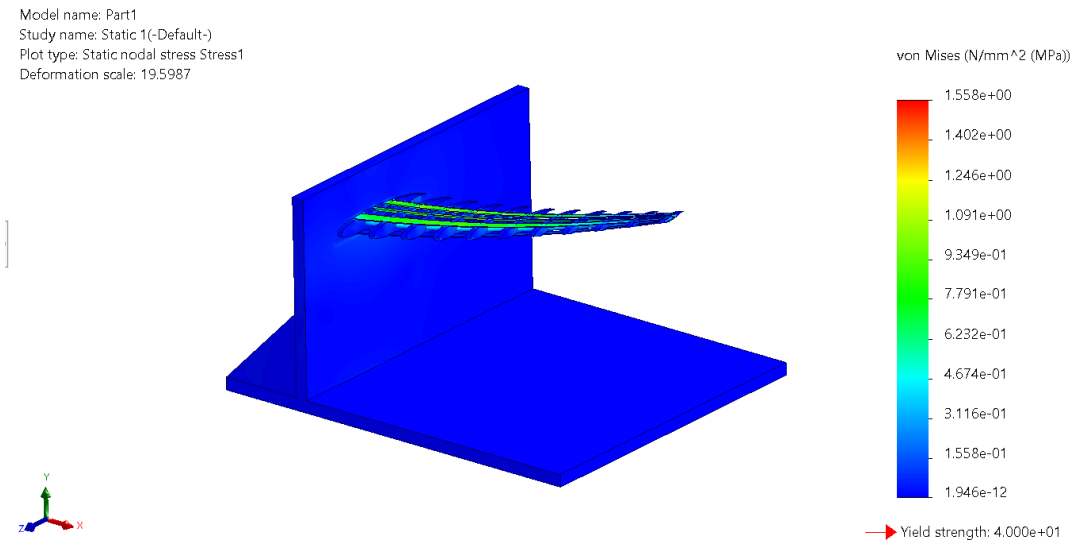


Figure 3. Stress Plot

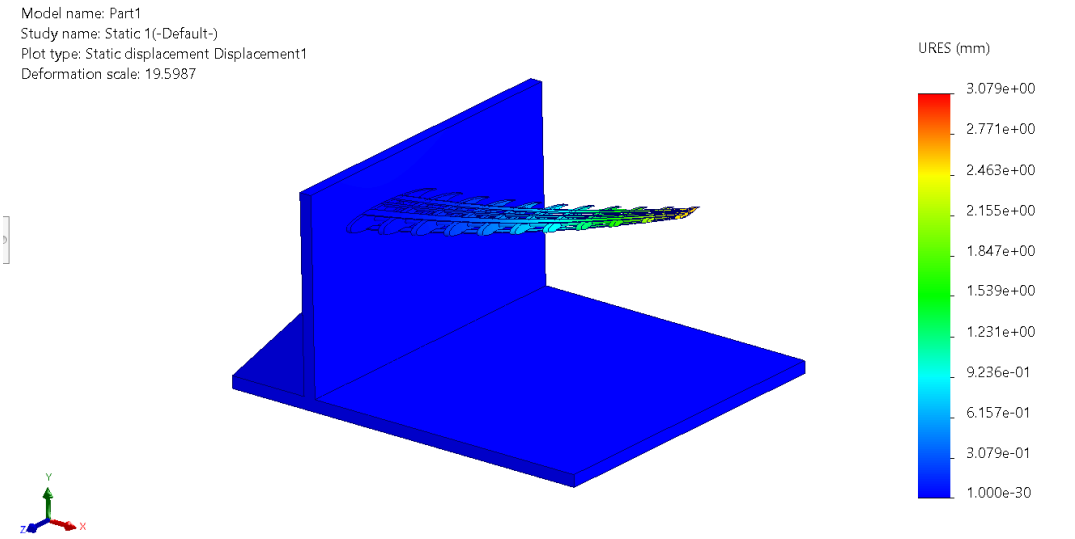


Figure 4. Displacement Plot

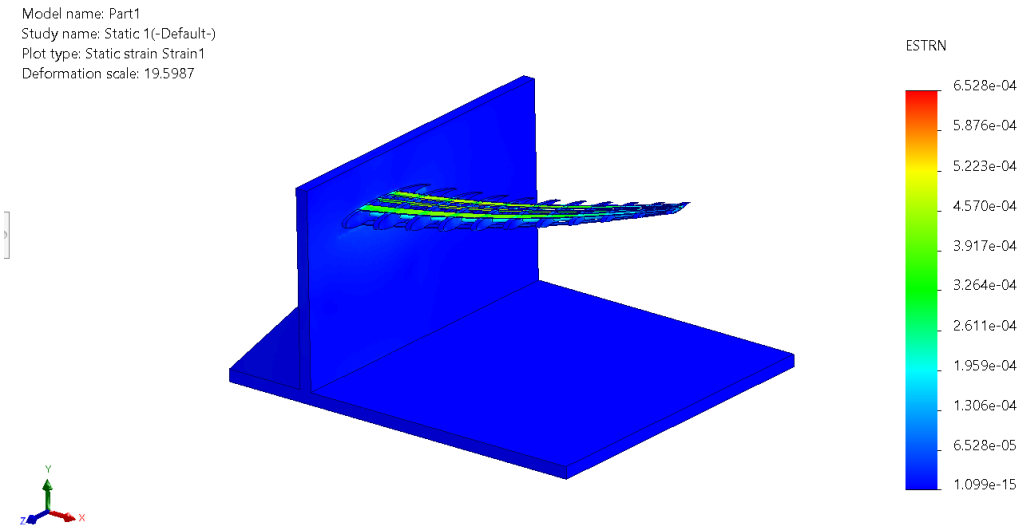


Figure 5. Strain Plot

Summary Results	
Max. Stress	1.55 MPa
Max. Deformation	3 mm
Max. Strain	0.00065

NOTHING FOLLOWS